Lesson plan · for the teacher

The middle and the spread

Data science · Level 1 · Lesson 2

Time
40 min

Materials
A tape measure or a metre ruler; Last test's marks (no names); A phone or tablet with Darsi, for the lab

Learning goals

- Work out the mean, median and mode.
- See how one extreme value pulls the mean.
- Measure the spread with the range.

Activity: A human number line

1. Warm-up (5 min): Ask: if a parent wants one number for the class's test, which number should we give, and why?
2. Teach (8 min): Work through nine marks on the board: total and share for the mean, order for the median, count for the mode, highest minus lowest for the range.
3. Measure (15 min): Nine volunteers measure their hand spans and line up in order. The middle pupil holds a card saying "median". The class works out the mean and the range.
4. Extreme (7 min): Add your own (much bigger) hand span. Recalculate: which number moved a lot, and which barely moved?
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence, with a number in it. Then give the exit ticket.

Discussion

- Why did the median barely move when the teacher joined?
- When is the mode the only middle you can use?
- What does a big range tell a teacher?

Exit ticket

- For 2, 4, 4 and 10, write the mean, the median, the mode and the range.

Try it in class tomorrow
Line up nine pupils by height: the fifth is the median. Then work out the mean and the range together on the board.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **A class has a range of 18 marks. What does that tell you?**

◯ Eighteen pupils passed ◯ The average is 18 ◯ The marks are very spread out

2 **Complete the sentences.**

For 3, 5 and 10, the _____ is 6 and the _____ is 5.

Word bank: range · median · mean · mode

3 **Nine pupils' marks out of 20 are in the array. What does this program print?**

```
from ulab import numpy as np
marks = np.array([9, 11, 12, 12, 13, 14, 14, 15, 16])
print(round(np.mean(marks), 1))
print(np.median(marks))
```

◯ 16, then 9 ◯ 13.0, then 12.9 ◯ 12.9, then 13.0 ◯ 116, then 13.0

4 **Pupils' journey times in minutes are 5, 6, 8, 9 and 52. Which number best describes a typical journey?**

◯ The median, 8 minutes ◯ The range, 47 minutes ◯ The mean, 16 minutes

5 **Match each question to the best summary.**

What is the most popular school lunch?	• • • •	The mean
What is a typical family's journey time, with a few very long ones?	• • • •	The mode
How much rice does the class eat per pupil, shared equally?	• • • •	The range
How different are the marks in this class?	• • • •	The median

Answers · for the teacher

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1 **The marks are very spread out**

The range is the highest mark minus the lowest: it measures spread, not the middle.

2 **For 3, 5 and 10, the mean is 6 and the median is 5.**

The total is 18 and there are 3 values, so the mean is 6.

3 **12.9, then 13.0**

The first line prints the mean: the marks add to 116, and 116 ÷ 9 is about 12.9.

4 **The median, 8 minutes**

One very long journey pulls the mean up to 16, longer than four of the five journeys.

5 **What is the most popular school lunch? — The mode**
What is a typical family's journey time, with a few very long ones? — The median
How much rice does the class eat per pupil, shared equally? — The mean
How different are the marks in this class? — The range

Categories need the mode, skewed data the median, sharing out the mean, and differences a measure of spread.

For you to think about

Which number do you usually report to parents after a test? Would the median or the range tell them something new?